BEFORE YOU READ

Twenty integrals in seven families

The integrals on this page group by the kind of integrand. Each row shows the integrand and its antiderivative; add + C to every one.

- L0 **Special integrals**
  - L1 **Power**
    - L2 x^n $n \neq -1$ — $x^{n+1} / (n+1)$
    - L2 $1 / x$ — $\ln |x|$
  - L1 **Exponential**
    - L2 e^x — e^x
    - L2 a^x $a > 0, a \neq 1$ — $a^x / \ln a$
    - L2 e^{kx} $k \neq 0$ — e^{kx} / k
  - L1 **Trigonometric**
    - L2 $\sin x$ — $-\cos x$
    - L2 $\cos x$ — $\sin x$
    - L2 $\sec^2 x$ — $\tan x$
    - L2 $\csc^2 x$ — $-\cot x$
    - L2 $\sec x \cdot \tan x$ — $\sec x$
    - L2 $\csc x \cdot \cot x$ — $-\csc x$
  - L1 **Inverse trigonometric**
    - L2 $1 / (1 + x^2)$ — $\arctan x$
    - L2 $1 / \sqrt{1 - x^2}$ — $\arcsin x$
    - L2 $1 / (x \cdot \sqrt{x^2 - 1})$ — $\operatorname{arcsec} |x|$
  - L1 **Logarithmic**
    - L2 $\ln x$ — $x \ln x - x$
  - L1 **Logarithmic pattern**
    - L2 $f'(x) / f(x)$ — $\ln |f(x)|$
  - L1 **Trigonometric to logarithm**
    - L2 $\tan x$ — $\ln |\sec x|$
    - L2 $\cot x$ — $\ln |\sin x|$
    - L2 $\sec x$ — $\ln |\sec x + \tan x|$
    - L2 $\csc x$ — $\ln |\csc x - \cot x|$

TAKEAWAY

Where the logarithms come from

Logarithms appear whenever the integrand has the form f′(x) / f(x): 1/x, tan x, cot x, sec x and csc x all follow that pattern.